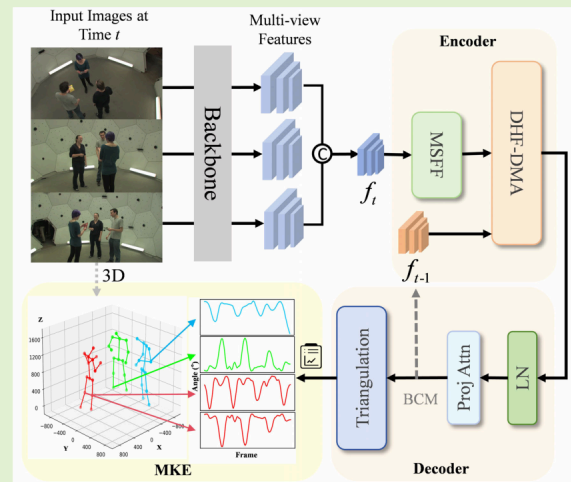


MoViSense: Multiview Spatiotemporal Transformer for 3-D Human Kinematics Sensing

Dezhi Zheng¹, Member, IEEE, Zhiyi Yang², Wenfeng Liu³, Xiao Liang⁴, Chun Hu⁵, and Kang Ma⁶

Abstract—Human pose estimation has wide applications in health monitoring, disease diagnosis, and motion rehabilitation. These applications rely on motion assessment using kinematic parameters, which can be obtained from the coordinates of human body key points. Most current noncontact key point measurement methods rely on multiple cameras to reconstruct the human body in realistic multiperson interaction and occlusion scenarios. However, sparse camera configurations often lead to limited pose estimation accuracy, while increasing the number of viewpoints to support high-accuracy kinematic analysis reduces efficiency. In this study, MoViSense is proposed for 3-D human pose estimation and kinematic analysis under sparse camera configurations by exploiting the spatial and temporal continuity. Based on the Transformer, the encoder integrates a multiscale gated feedforward (MSFF) module to enhance spatial representations and cross-view alignment, while a dynamic history fusion-deformable multiscale attention (DHF-DMA) module utilizes temporal continuity of human motion to improve robustness under occlusion. In addition, a biomechanical constraint mechanism (BCM) enforces bone-length consistency. A motion kinetics extractor (MKE) converts estimated 3-D key points into interpretable kinematic parameters. Experiments on the CMU Panoptic dataset show that MoViSense achieves an AP50 of 86.49 and an MPJPE of 27.85 mm, outperforming the other representative methods under sparse camera configurations. The relative deviation (RD) of stride length was 5.37%, and the RD of cadence was 1.45%.

Index Terms—Kinematic analysis, motion assessment, multiview 3-D human pose estimation, Transformer.



I. INTRODUCTION

HUMAN pose estimation is an important research topic in intelligent perception and kinematic analysis. It has been widely applied in health monitoring, disease diagnosis, and

motion rehabilitation [1], [2], [3]. These applications mainly rely on motion assessment using kinematic parameters such as joint angles, step length, cadence, and symmetry index (SI), which can be obtained from the coordinates of human body key points. Previous studies have shown that these parameters are highly correlated with clinical manifestations of diseases such as stroke, Parkinson's disease, and musculoskeletal disorders [1], [2], [3], [4], [5], [6].

Traditional motion assessments mainly rely on motor function assessment scales, such as MDS-UPDRS [7] and FMA-LE [8], which subjectively score the motor abilities of patients and lack repeatability and long-term monitoring capability. To achieve objective and quantitative motion assessment, current sensing approaches are mainly divided into two categories: contact systems and noncontact visual systems. Contact systems based on inertial measurement units (IMUs) or surface electromyography (sEMG) can obtain high-precision measurements, but they require complex wearing procedures and calibration, which limits their practical applicability [5]. In contrast, noncontact visual measurement

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Dezhi Zheng, Zhiyi Yang, Wenfeng Liu, Xiao Liang, and Chun Hu are with the School of Interdisciplinary Sciences, the School of Information and Electronics, the State Key Laboratory of Environment Characteristics and Effects for Near-Space, the State Key Laboratory of CNS/ATM, and the MIT Key Laboratory of Complex-Field Intelligent Exploration, Beijing Institute of Technology, Beijing 100081, China (e-mail: zhengdezhi@bit.edu.cn; zzy10814@163.com; wflui@bit.edu.cn; liangxiao@bit.edu.cn; bit_hc@bit.edu.cn).

Kang Ma is with the School of Computer Science and Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: makang@bit.edu.cn).

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methods have the advantages of flexibility and high precision [3], [4], [9]. In particular, RGB cameras are characterized by low cost and ease of deployment, making them a practical solution for long-term health monitoring in community and home environments [3], [4]. However, many existing RGB camera systems rely on densely deployed cameras to obtain more viewpoints. Additional viewpoints can enhance pose estimation performance but also increase hardware and computational costs, which weakens the advantages of RGB solutions.

In fact, the number and spatial arrangement of cameras together determine the strength of geometric constraints, which affects the accuracy of pose estimation. Specifically, existing methods can be divided into monocular and multiview methods. Monocular methods extend pose estimation to 3-D space through volume prediction [10] or coordinate regression [11], [12]. Although they have the advantages of low cost and easy deployment, they suffer from significant depth ambiguity, resulting in limited accuracy [13]. Moreover, the number of detected key points can be insufficient to reliably identify the human body in real scenarios with multiperson interactions and occlusions. This leads to missed detections and fails to stably construct 3-D key point estimation, which further affects the calculation of kinematic parameters [9], [13], [14]. In contrast, multiview methods reduce depth ambiguity by integrating information from different viewpoints, which enhances the accuracy and robustness of pose estimation in real scenarios [15]. For instance, fast and robust MP-3-D [16] reconstructs individual skeletons by cross-view key points clustering, achieving high accuracy in complex scenarios. However, existing high-performance methods often rely on the deployment of dense camera configurations, which require high hardware costs and complex calibration procedures, limiting the flexibility of the system.

Furthermore, the Transformer presents strong advantages in global feature perception. Therefore, most representative methods combine them with the local feature extraction ability of convolutional networks to improve the accuracy of 3-D key point estimation [17], [18], [19]. For instance, direct MvP [20] and MVGFormer [21] have demonstrated excellent performance in high-resolution and multiperson scenarios. However, these methods mainly adopt the frame-by-frame reasoning approach, which ignores the inherent temporal continuity of human motion and the smooth characteristics of key points in the temporal dimension [22]. Especially under occlusion, the model cannot effectively infer missing key points, which degrades the stability and accuracy of pose estimation [23]. Therefore, some studies have begun to introduce temporal modeling mechanisms. For example, KTPFormer [24] embeds the kinematic topology and trajectory priors into the attention calculation process to enhance the modeling ability of cross-frame correlations. However, it has high requirements for the number of cameras and the spatial layout. Specifically, the observed features are missing under sparse view scenarios, which makes it difficult to estimate the coordinates of human body key points, resulting in structural errors such as inconsistent bone lengths and abnormal limb proportions. It directly affects the reliability and stability of the motion

assessment results, making it difficult to convert the 3-D key point sequence into clinically significant kinematic indicators.

This study proposes MoViSense, a multiperson multiview 3-D pose estimation and motion analysis framework for sparse view configurations of three cameras. This framework combines camera configuration design with a Transformer-based architecture. Through spatial feature enhancement, temporal continuity utilization, and human body structure prior constraints, it achieves high-precision pose estimation under sparse camera configurations. In addition, a kinematic parameter extraction module is introduced to convert the 3-D key point sequence of pose estimation into interpretable kinematic parameters, which constructs a systematic pipeline from pose estimation to motion analysis.

The main contributions of this study are summarized as follows.

- 1) *Integrated Spatiotemporal Fusion Encoder*: A spatiotemporal feature fusion encoder for sparse camera configurations was proposed. The multiscale gated feed-forward (MSFF) module enhances the expression ability of human structure through local-global feature fusion. The dynamic historical fusion-deformable multiscale attention (DHF-DMA) module utilizes the interframe continuity of human motion and introduces the features of the previous moment to guide the current moment's pose estimation. It effectively alleviates the problems of depth blurring and occlusion, achieving 4.42 higher in average precision at 50 mm (AP50) and a reduction of 5.45 mm in mean per joint position error (MPJPE).
- 2) *Biomechanical Constraint Learning*: The biomechanical constraint mechanism (BCM) incorporates bone-length consistency as an explicit skeletal prior, improving the structural consistency and robustness of human pose estimation.
- 3) *Clinically Interpretable Kinematic Extraction*: The motion kinetics extractor (MKE) is introduced to generate the 3-D key point sequences of pose estimation. It converts 3-D key point sequences into stable kinematic parameters with smoothing and filtering, which supports interpretable motion analysis under camera configurations.

The remainder of this study is organized as follows. Section II describes the proposed methodology in detail. Section III presents the experimental results. Section IV concludes this study.

II. METHODS

A. Overall Structure

The proposed MoViSense is a multiperson multiview 3-D human pose estimation and kinematic analysis framework designed for a three-camera configuration. The framework aims to transform multiview image features into a unified 3-D key points representation, which achieves high-precision pose reconstruction and interpretable motion assessment under sparse camera configurations.

As illustrated in Fig. 1, MoViSense consists of four major components: an image backbone for multiview feature

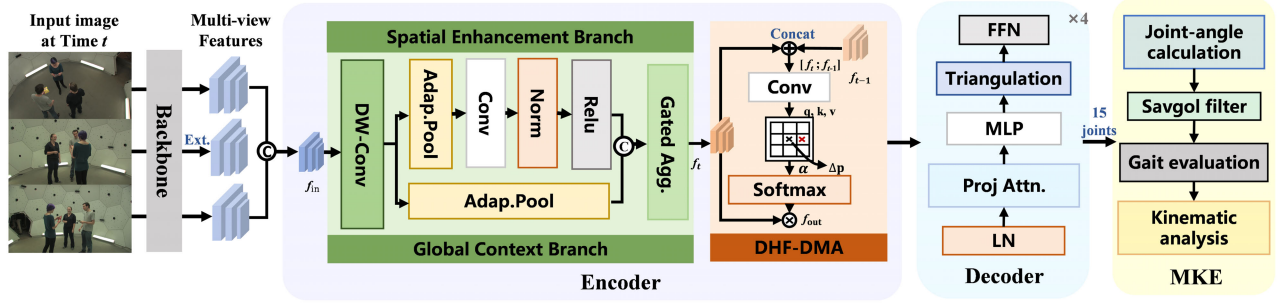


Fig. 1. Framework of the proposed MoViSense system.

extraction; an MSFF module for spatial enhancement; a DHF-DMA module for temporal modeling and cross-view alignment; and an MKE that converts estimated 3-D key point sequences into clinically interpretable kinematic parameters. In addition, a BCM is embedded during network optimization to enforce bone-length consistency, ensuring biomechanically realistic reconstruction.

In the processing pipeline, spatial features are first extracted from synchronized multiview images by the backbone network and refined through the MSFF module to capture fine-grained structural details. These enhanced features are then fed into the DHF-DMA module, which integrates temporal information from historical frames and achieves adaptive spatiotemporal fusion across camera views. Finally, the reconstructed 3-D joints are processed by the MKE to compute quantitative gait and motion indicators such as joint-angle curves, stride length, cadence, and SI. Through this design, MoViSense establishes a unified pipeline from multiview perception to biomechanical interpretation.

B. Encoder Module

The encoder module is composed of two core submodules: the MSFF module and the DHF-DMA module.

1) *Multiscale Feedforward Module*: In multiview multiperson 3-D human pose estimation, most traditional methods only rely on local convolutional features or global feature aggregation. These methods are susceptible to occlusion and scale variations, leading to the loss of joint structure information, which becomes more severe under sparse camera configurations. Therefore, this study designs an MSFF module, aiming to enhance the representation capability of the feature space through multiscale convolution and gating mechanisms (Fig. 2). MSFF employs a dual-branch local–global feature fusion strategy, by simultaneously modeling local spatial details and global semantic context, obtaining more robust spatial representations.

Specifically, the input feature f_{in} at time t is obtained by concatenating the feature maps extracted from the multiview images captured by N cameras using the backbone network. After linear transformation and depthwise separable convolution, the features are divided into two branches: the spatial enhancement branch and the global context branch. The former extracts local spatial features through pooling, convolution, normalization, and activation operations, which enhances the spatial continuity and texture discriminability of the joint

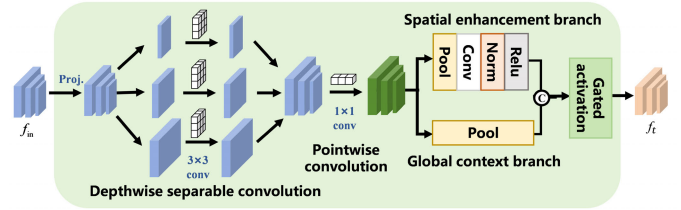


Fig. 2. Architecture of the MSFF module.

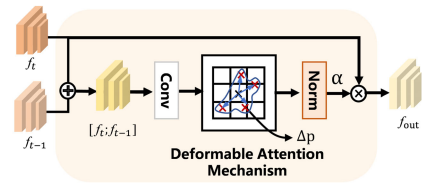


Fig. 3. Architecture of the DHF-DMA module.

neighborhood. While the latter integrates global semantic information through adaptive pooling and upsampling, which introduces global feature understanding of the overall human structure layout and cross-view consistency. Finally, the outputs of these two branches are fused in the channel dimension, and the gated activation mechanism dynamically balances the contributions of local and global information to generate the spatially adaptive output feature f_t .

MSFF explicitly integrates local details and global context under sparse camera configurations, which provides higher quality input features for the subsequent DHF-DMA cross-view alignment.

2) *DHF-DMA Module*: The traditional pose estimation methods lack the utilization of temporal information regarding the interframe relationships of the 3-D key points sequence. Based on the continuity of human movement, this study designs a DHF module, aiming to dynamically fuse the historical feature f_{t-1} and the current feature f_t to achieve cross-frame temporal information modeling (Fig. 3). The DMA module combines deformable convolution with a multiscale attention mechanism to perform feature sampling and adaptive alignment. It establishes temporal correspondence by utilizing historical features from time $t - 1$ to guide attention computation at time t , which dynamically aggregates cross-frame information using learnable sampling offsets and attention weights to generate the refined feature f_{out} .

The formulation can be expressed as follows:

$$O_t = \text{DHF}([F_t; F_{t-1}]) + F_t \quad (1)$$

where O_t represents the fused output at time t , F_t denotes the current feature, and F_{t-1} represents the historical feature from the previous frame. The deformable sampling process of the DMA module is formulated as follows:

$$\text{DMA}(X) = \sum_{l=1}^L \sum_{h=1}^H \sum_{p=1}^P \alpha_{l,h,p} X(p + \Delta p_{l,h,p}) \quad (2)$$

of which the offset $\Delta p_{l,h,p}$ is jointly determined by $[F_t; F_{t-1}]$. The vertical concatenation of current feature F_t and historical feature F_{t-1} enables the sampling points to adaptively align with the semantic centers of the historical frame and model the temporal dependencies.

Specifically, when $t = 0$, there is no historical feature available, and $[F_t; F_{t-1}]$ degenerates into $[F_t; F_t]$, which simplifies the operation to a self-attention mechanism without temporal information.

The attention weight $\alpha_{l,h,p}$ is normalized through the softmax function to achieve weighted aggregation within the same attention head, ensuring both interpretability and stability of the attention distribution.

C. Decoder Module

To improve compatibility with the encoder's temporal guidance and enhance spatial structure reconstruction, a layer normalization layer is incorporated into the decoder to normalize feature representations and facilitate subsequent information processing in the project attention module.

In each decoder layer, pose queries are initialized at the human body joint level, and a spatial attention mechanism is employed to query and aggregate spatial features from multiple camera views. The aggregated features are then fed into the triangulation module, followed by a feedforward neural network to predict the 3-D key point and their corresponding confidence scores. The predicted results are propagated as inputs to the next decoder layer. After stacking N decoder layers, the final high-precision 3-D key points are obtained and then fed into the joint-angle computation module for further motion analysis.

D. Biomechanical Constraint Mechanism

To further improve the accuracy of 3-D human pose estimation, a BCM is designed by incorporating the structural priors of the human skeleton. The proposed method enforces consistency of predicted poses by constraining the relative ratios between individual bone lengths and their average lengths. This biomechanical prior enables the model to reconstruct structurally plausible 3-D poses more stably, particularly when certain joints are occluded or affected by noise. The BCM is implemented through the bone-length consistency loss, which encourages biomechanically plausible pose predictions.

The overall training objective consists of two components: the coordinate regression loss L_{reg} and the bone-length consistency loss L_{bone} . The total loss is defined as follows:

$$L_{\text{total}} = L_{\text{reg}} + \lambda L_{\text{BCL}} \quad (3)$$

where λ is a balancing coefficient that controls the relative contribution of the bone-length consistency constraint. The computation begins with the calculation of bone vectors as follows:

$$v_{\text{pred}}^{(b,n)} = J_{\text{pred}}^{(b,i_n)} - J_{\text{pred}}^{(b,j_n)} \quad (4)$$

$$v_{\text{gt}}^{(b,n)} = J_{\text{gt}}^{(b,i_n)} - J_{\text{gt}}^{(b,j_n)} \quad (5)$$

where $J_{\text{pred}}^{(b,j)} \in \mathbb{R}^3$ and $J_{\text{gt}}^{(b,j)} \in \mathbb{R}^3$ denote the predicted and ground-truth 3-D joints, respectively. $v_{\text{pred}}^{(b,n)} \in \mathbb{R}^3$ and $v_{\text{gt}}^{(b,n)} \in \mathbb{R}^3$ denote vectors.

Subsequently, the bone lengths are derived as follows:

$$\ell_{\text{pred}}^{(b,n)} = \|v_{\text{pred}}^{(b,n)}\|_2 \quad (6)$$

$$\ell_{\text{gt}}^{(b,n)} = \|v_{\text{gt}}^{(b,n)}\|_2. \quad (7)$$

Based on these bone lengths, the relative bone-length ratios are computed as follows:

$$r_{\text{pred}}^{(b,n)} = \frac{\ell_{\text{pred}}^{(b,n)}}{\bar{\ell}_{\text{pred}}^{(b)} + \varepsilon} \quad (8)$$

$$r_{\text{gt}}^{(b,n)} = \frac{\ell_{\text{gt}}^{(b,n)}}{\bar{\ell}_{\text{gt}}^{(b)} + \varepsilon} \quad (9)$$

where $\bar{\ell}_{\text{gt}}^{(b)} = 1/N \sum_{n=1}^N \ell_{\text{gt}}^{(b,n)}$ represents the mean bone length of the current frame and $\varepsilon = 1e^{-6}$ is a small constant to prevent numerical instability. The bone-length consistency loss L_{BCL} is finally formulated as:

$$L_{\text{BCL}} = \frac{1}{BN} \sum_{b=1}^B \sum_{n=1}^N w^{(b,n)} |r_{\text{pred}}^{(b,n)} - r_{\text{gt}}^{(b,n)}|. \quad (10)$$

E. Motion Kinetics Extractor

The MKE module is introduced to convert estimated 3-D key points into clinically interpretable kinematic parameters for motion assessment. It addresses the limitation of conventional pose estimation methods that only output coordinates without motion interpretation, allowing quantitative indicators for rehabilitation assessment, gait analysis, and motor disorder detection.

The human skeletal structure and joint indexing order are predefined as illustrated in Fig. 4, ensuring consistency in angle and temporal parameter computation. Each marker denotes a key point extracted from multiview pose estimation and connected according to anatomical linkage. A 3-D skeleton is constructed using 15 key joints to support joint-angle computation and posture constraints. All experimental results are derived from a single-subject walking sequence of 140 frames in the dataset, ensuring temporal coherence among joint angles and kinematic parameters. This sequence is used to evaluate the dynamic performance of the MKE module during continuous motion.

The MKE module takes joint coordinates estimated by the 3-D human pose estimation network as input, incorporating auxiliary information such as individual height and video frame rate to extract key gait parameters and analyze bilateral

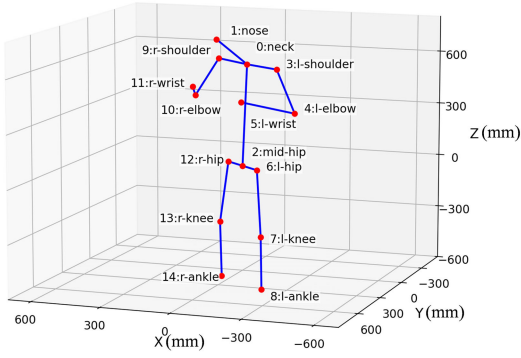


Fig. 4. 3-D human skeleton joint labeling scheme used in the proposed MKE module.

symmetry. The joint-angle θ formed by three adjacent points A , B , and C is computed as follows:

$$\theta_{ABC} = \arccos \left(\frac{\vec{AB} \cdot \vec{CB}}{\|\vec{AB}\| \cdot \|\vec{CB}\|} \right) \quad (11)$$

where A , B , and C denote the 3-D coordinates of adjacent joints. θ is the joint angle.

During temporal analysis, gait cycles T_g are segmented by detecting local extrema of the vertical ankle displacement $Z_{\text{ankle}}(t)$, from which step counts and stride lengths are obtained. The stride length is calculated as follows:

$$L_{\text{stride}} = \left| x_{\text{ankle}}^{(n+2)} - x_{\text{ankle}}^{(n)} \right| \quad (12)$$

where $x_{\text{ankle}}^{(n)}$ represents the horizontal position of the n th heel strike on the same side. The step frequency is computed as follows:

$$\text{Cadence} = \frac{N_{\text{steps}}}{t_N - t_1} \times 60 \quad (13)$$

where N_{steps} is the total number of steps. t_1 and t_N are the time instants of the first and last detected heel strikes, respectively. Based on the ratio between the stance and swing phases, the stance-to-swing ratio is derived as follows:

$$R_{\text{stance/swing}} = \frac{N_{\text{stance}}}{N_{\text{swing}}} \quad (14)$$

which also reflects stability when evaluated together with the vertical displacement of the contralateral foot at heel strike moments.

To characterize bilateral coordination, the commonly used clinical metric SI is introduced

$$\text{SI} = \frac{|L - R|}{0.5(|L| + |R|)} \times 100\% \quad (15)$$

where L and R denote the mean joint angles of the left and right sides, respectively. A smaller SI value represents better bilateral motion symmetry.

Furthermore, the Hilbert transform is applied to extract the instantaneous phase of joint-angle signals $\phi_L(t)$ and $\phi_R(t)$, and the mean phase difference is computed as follows:

$$\Delta\phi_{\text{mean}} = \frac{1}{T} \int_0^T |\phi_L(t) - \phi_R(t)| dt. \quad (16)$$

To quantify temporal consistency between the two sides, the signal-to-noise ratio (SNR) is calculated as the ratio of signal variance to error variance

$$\text{SNR} = 10 \log_{10} \left(\frac{\text{Var}(\theta_{\text{signal}})}{\text{Var}(\theta_{\text{error}})} \right) \quad (17)$$

which evaluates the smoothness and reconstruction stability of the output curves.

The MKE module extracts these kinematic features to generate clinically interpretable kinematic parameters, such as joint angles, step length, cadence, and SI. This module provides an economic and clinically interpretable method for gait monitoring, motion assessment, and long-term visual tracking in rehabilitation applications.

Sections III presents comprehensive quantitative and qualitative evaluations of the proposed MoViSense framework under sparse camera configurations.

III. EXPERIMENTAL RESULTS

A. Experimental Setup

The proposed MoViSense framework was evaluated on the CMU Panoptic dataset [25], one of the largest multiview 3-D human pose benchmarks. It consists of 31 synchronized HD cameras (1920×1080 , 30 fps) arranged in a hemispherical layout, providing comprehensive view coverage. The dataset was captured in a controlled indoor environment with uniform lighting. Moreover, this dataset contains over 3 h of multiperson interaction sequences, with up to eight people simultaneously active in a scene. These sequences involve significant depth ambiguity and frequent occlusions, making the dataset a challenging benchmark for multiperson 3-D pose estimation.

The dataset adopts MPJPE and AP50 as quantitative evaluation metrics. Specifically, MPJPE represents the Euclidean distance between predicted and ground-truth joint coordinates; AP50 represents the average precision under the 50-mm threshold, reflecting detection precision.

Cross-dataset experiments on the Shelf and Campus datasets were conducted to evaluate the generalization ability of the proposed method under different environments and camera configurations. The Shelf dataset is an indoor multiperson dataset containing five cameras, with severe occlusions among subjects; the Campus dataset is an outdoor three-camera dataset with significant lighting changes and background clutter. These two datasets differ from Panoptic in terms of environment, camera configurations, and subject movement patterns, making them suitable for evaluating cross-scenario robustness under sparse view conditions. For Shelf and Campus, we follow the official evaluation protocol using the Percentage of Correct Parts (PCP). PCP measures whether predicted body segments match the ground truth within a predefined threshold. Results were reported for the three standard test subsequences A1, A2, and A3, together with their average performance.

All experiments were conducted on a workstation equipped with eight NVIDIA GeForce RTX 3090 GPUs (24 GB each). Following previous benchmarks [20], [26], the ResNet-50

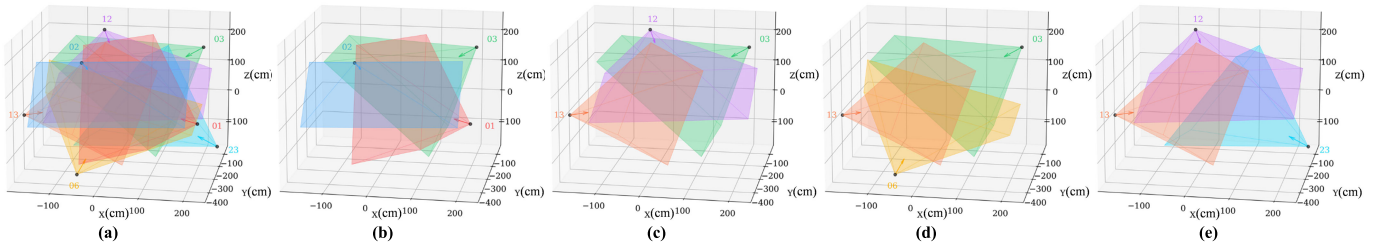


Fig. 5. 3-D spatial distribution and field-of-view coverage of the Panoptic camera setup. (a) All cameras. (b) Cam_01–Cam_02–Cam_03. (c) Cam_03–Cam_12–Cam_13. (d) Cam_03–Cam_06–Cam_13. (e) Cam_12–Cam_13–Cam_23.

backbone was employed for 2-D feature extraction. The network was pretrained on the COCO key point dataset [27], and subsequently fixed during 3-D model training to stabilize feature representation. The proposed framework was trained using the Adam optimizer with an initial learning rate of $1e^{-4}$ and 100 epochs.

B. Camera Layout and View Number Analysis

In the multiview 3-D human pose estimation system, the spatial arrangement of the cameras directly determines the effective observation range under sparse camera configurations, which affects the stability and accuracy of the measurement results. According to the triangulation error propagation theory [28], [29], [30], the positions of 3-D key points are determined by the intersection of multiple lines of sight. When the lines of sight approach parallelism, the 2-D positioning error will be amplified in the depth direction. In contrast, the triangulation will become more stable when the angle is within the reasonable range of 40° – 140° . Therefore, a reasonable spatial layout is important in 3-D reconstruction accuracy when the number of cameras is limited.

Based on the camera intrinsic and extrinsic parameters of the Panoptic dataset, geometric metrics, including mean baseline, mean viewing angle, and triangle area of each camera triplet, were computed to quantify spatial distribution and perspective expansion. Specifically, the mean baseline is the average distance between camera pairs; the mean viewing angle is the average angle between their optical axes; and the triangle area is formed by the three-camera centers. The camera combinations were ranked using a normalized geometric score to determine the optimal three-camera layout. Fig. 5 presents the visual results of the 3-D spatial distribution and field of view coverage of different camera combinations in the Panoptic setup.

As listed in Table I, there were significant performance differences among camera combinations. The combinations with a more concentrated view distribution (Cam_01–Cam_02–Cam_03) performed worse on AP50 and MPJPE, confirming that a limited baseline and overlapping coverage would weaken the depth constraint ability. In contrast, the combinations with a more balanced spatial distribution (Cam_03–Cam_06–Cam_13) significantly improved the accuracy of 3-D reconstruction. When the camera distribution (Cam_12–Cam_13–Cam_23) was further optimized with larger expansion angles in both the horizontal and vertical directions, the model performance achieved the optimal level.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT THREE-CAMERA COMBINATIONS

Camera Triplet	Mean Baseline (cm)	Mean Viewing Angle($^\circ$)	Triangle Area (cm^2)	AP50	MPJPE (mm)
Cam_01–Cam_02–Cam_03	294.56	75.08	36289.81	54.19	42.45
Cam_03–Cam_12–Cam_13	342.24	92.99	36113.14	77.60	31.23
Cam_03–Cam_06–Cam_13	389.11	114.90	50243.35	81.10	29.07
Cam_12–Cam_13–Cam_23	424.80	118.26	71333.97	86.49	27.85

TABLE II

PERFORMANCE COMPARISON WITH DIFFERENT NUMBERS OF CAMERAS

Views	Configuration	AP50 $\uparrow$	MPJPE(mm) $\downarrow$
2	Cam_13–Cam_23	14.13	100.48
3	Cam_12–Cam_13–Cam_23	86.49	27.85
4	Cam_12–Cam_13–Cam_16–Cam_23	87.57	26.64
5	Cam_3–Cam_12–Cam_13–Cam_16–Cam_23	95.56	21.01

Therefore, Cam_12–Cam_13–Cam_23 was adopted as the default three-camera configuration in the experiments.

Table II provided that the performance deteriorated significantly when only two cameras were used, revealing that this configuration was difficult to recover the 3-D structure under occlusion. Increasing the number of cameras to three significantly reduced MPJPE. Further increasing the number of cameras to four or five improved accuracy but also increased computational overhead and reduced efficiency.

Therefore, the three-camera configuration reached a better balance between accuracy and practical deployment.

C. Quantitative Comparison With State-of-the-Art Methods

To validate the effectiveness of the proposed framework, MoViSense was quantitatively compared with several state-of-the-art 3-D human pose estimation methods, including geometry-based approaches (e.g., MvP [20] and Fast VoxelPose [31]) and Transformer-based frameworks (e.g., MVGFormer [21]).

As listed in Table III, MoViSense achieved the highest AP50 (86.49) and the lowest MPJPE (27.85 mm) among all evaluated methods. Compared with the latest strong baseline

TABLE III

COMPARISON WITH STATE-OF-THE-ART MULTI-VIEW 3-D POSE ESTIMATION METHODS

Method	Venue	AP50 $\uparrow$	MPJPE(mm) $\downarrow$
MvP [20]	NeurIPS 21	73.84	34.94
Fast VoxelPose [31]	ECCV 22	72.67	34.77
MVGFormer [21]	CVPR 24	82.07	33.30
MoViSense	Ours	86.49	27.85

TABLE IV

ABLATION RESULTS FOR THE PROPOSED ARCHITECTURE

MSFF	DHF-DMA	BCM	AP50 $\uparrow$	MPJPE(mm) $\downarrow$
$\times$	$\times$	$\checkmark$	80.95	32.02
$\checkmark$	$\times$	$\times$	85.51	31.41
$\times$	$\checkmark$	$\times$	80.89	30.92
$\checkmark$	$\checkmark$	$\times$	84.86	28.87
$\checkmark$	$\checkmark$	$\checkmark$	86.49	27.85

MVGFormer [21], MoViSense achieved an AP50 improvement of 4.42 and reduced MPJPE by 5.45 mm, highlighting the advantages of the proposed modules. The results show that the proposed approach can achieve competitive 3-D human pose estimation accuracy, with an economic configuration using only three cameras.

D. Ablation Study

To evaluate the contribution of each component to the model performance, an ablation study was conducted on three key modules: MSFF, DHF-DMA, and BCM. All experiments were performed under identical training and testing protocols.

As provided in Table IV, the results demonstrated that introducing MSFF alone increased AP50 to 85.51 and reduced MPJPE to 31.41 mm, validating that multiscale spatial enhancement can improve local details and the performance of human pose estimation. When only DHF-DMA was introduced, MPJPE decreased to 30.92 mm, confirming that temporal continuity modeling can enhance the stability of 3-D reconstruction. When both MSFF and DHF-DMA are enabled simultaneously, MPJPE further decreased to 28.87 mm, showing the complementary effect of spatiotemporal fusion.

Furthermore, incorporating BCM achieved the best overall performance, confirming that bone-length consistency constraints can further suppress structural drift and enhance the stability of the pose sequence.

E. Cross-Dataset Generalization Under Sparse Camera Configurations

To further evaluate the generalization ability of the proposed method under different scenarios, cross-dataset experiments were conducted on two widely used multiview datasets, Shelf and Campus. Detailed descriptions of these datasets are provided in Section III-A. Two experimental settings were set: w/o fine-tune: the model was trained only on the Panoptic

TABLE V

CROSS-DATASET GENERALIZATION RESULTS ON SHELF AND CAMPUS

Datasets		Shelf				Campus			
Training Setting	Methods	A1	A2	A3	Average	A1	A2	A3	Average
w/o Finetune	MvP[20]	0.22	0	0.25	0.15	0	0	0.07	0.02
	Fast VoxelPose[31]	0	0	0	0	0	0	0	0
	MVGFormer[21]	88.42	78.38	87.70	84.83	41.43	58.84	70.14	56.80
	Ours	87.74	85.41	88.20	87.12	41.02	63.02	68.26	57.43
w/ Finetune	MvP	93.69	91.62	97.52	94.28	87.55	91.27	71.59	83.47
	Fast VoxelPose	99.25	95.14	97.64	97.34	96.53	94.39	97.46	96.13
	MVGFormer	99.18	95.95	97.76	97.63	92.04	90.63	90.94	91.21
	Ours	99.64	96.22	98.01	97.96	99.18	92.01	97.97	96.39

dataset and tested directly on the target dataset; w/ fine-tune: the model was fine-tuned on the target dataset before being tested.

As listed in Table V, without fine-tuning, geometry approaches exhibited severe performance degradation on both the Shelf and Campus datasets. Particularly, the PCP value was close to zero on the Campus dataset, demonstrating that the model had a strong dependence on camera parameters and poor generalization ability. In contrast, the Transformer-based model demonstrated better cross-domain stability.

After target domain finetuning, the performance of all methods significantly improved. MoViSense achieved an average PCP of 97.96% on Shelf and 96.39% on Campus. This demonstrated that the proposed spatiotemporal fusion method and BCM can effectively enhance the cross-domain stability of the dataset. It also verified the generalization ability of the model in the case of sparse viewpoint deployment.

F. Occlusion Subsequence Analysis

To evaluate the robustness under different occlusion conditions, occlusion-level subsequences were constructed on the CMU Panoptic dataset and quantified in a frame-by-frame manner. The ground-truth 3-D poses were projected onto Cam_23, one of the cameras in the optimized three-camera configuration. Joint occlusions were identified according to 2-D overlap and depth relationships. The number of occluded key points was statistically analyzed across the test set to determine occlusion thresholds. The 50th and 90th percentiles of the occluded joint distribution were used as thresholds, corresponding to two and eight occluded joints, respectively. Sequences with more than two occluded joints ($\text{Occl} \geq 2$) were defined as moderate occlusion subsequences, while sequences with more than eight occluded joints ($\text{Occl} \geq 8$) were defined as severe occlusion subsequences. The original referred to the entire test set without occlusion filtering. Table VI provides the evaluation results of each method under different occlusion levels.

The experimental results and visualizations (Fig. 6) showed that the performance of all methods decreased as occlusion

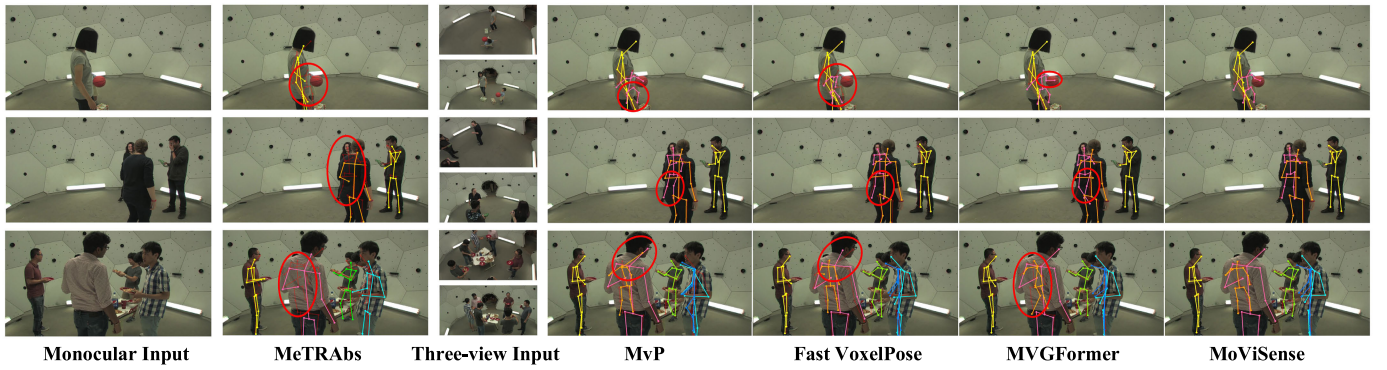


Fig. 6. Qualitative comparison of 3-D human pose estimation results under severe occlusion. (The red skeleton denotes the ground truth. The red circle represents abnormal predictions or structural distortions).

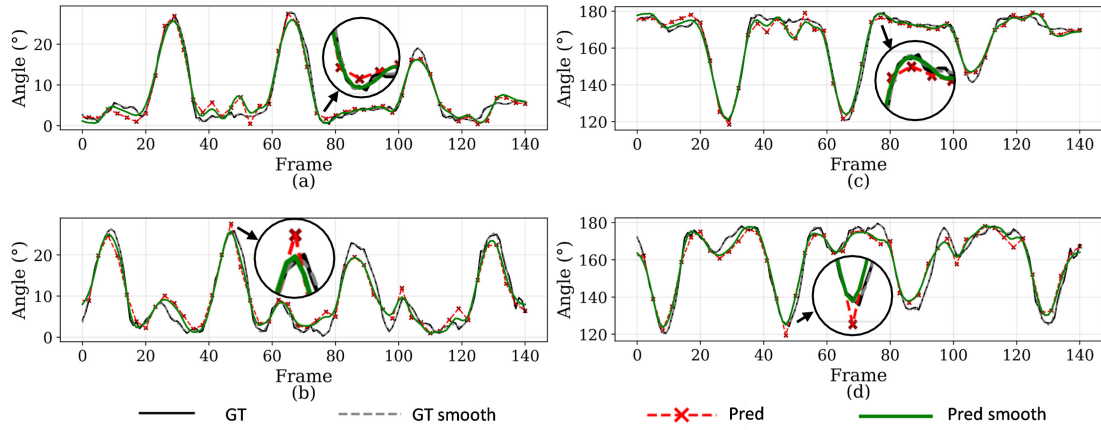


Fig. 7. Knee and ankle flexion-extension angle curves across gait cycles generated by the MKE module. (a) Left ankle flexion. (b) Right ankle flexion. (c) Left knee flexion. (d) Right knee flexion.

TABLE VI

PERFORMANCE COMPARISON OF SUBSEQUENCES WITH DIFFERENT DEGREES OF OCCLUSION

Methods	Original		Occl ≥ 2		Occl ≥ 8	
	AP50 $\uparrow$	MPJPE (mm) $\downarrow$	AP50 $\uparrow$	MPJPE (mm) $\downarrow$	AP50 $\uparrow$	MPJPE (mm) $\downarrow$
MeTRABs[13]	0	186.52	0	197.69	0	223.97
MvP	73.84	34.94	73.24	35.10	60.69	44.14
Fast VoxelPose	72.67	34.77	69.61	38.73	63.21	50.30
MVGFormer	82.07	33.30	80.99	34.08	72.95	43.37
MoViSense	86.49	27.85	85.51	28.46	77.40	35.45

became more severe, although the degree of degradation differed across methods. The monocular method MeTRABs [13] exhibited frequent missed detections under occlusion since severe visibility loss made human localization and 3-D reconstruction unreliable, resulting in an AP50 close to 0.

In contrast, multiview approaches demonstrated greater stability. However, their MPJPE still increased as occlusion became more severe. The proposed MoViSense maintained stable performance under moderate occlusion, with only a slight increase in MPJPE. Even under severe occlusion

sequences, it still maintained lower errors compared with other multiview methods. In severely occluded conditions, the proposed method better maintained the integrity of the skeleton topology and alleviated abnormal predictions and structural deformations.

G. Qualitative and Kinematic Evaluation

To verify the effectiveness of MoViSense in actual motion analysis tasks, qualitative visualization of the estimation results was performed, and kinematic parameters were evaluated. Fig. 7 provides a visual demonstration of a 140-frame walking sequence from the dataset [25]. The 3-D joint coordinates estimated by MoViSense were processed by the MKE module to generate flexion-extension angle trajectories for both knees and ankles. The black and gray curves represented the original and smoothed ground-truth angles (GT and GT smooth), respectively. The red dashed line and green solid line denote the predicted angles and the smoothed predicted angles (Pred and Pred smooth), respectively. As shown in the figure, the predicted curve generated by the MKE module can effectively track the variation trend of the ground-truth angle throughout the gait cycle. It maintained a relatively stable amplitude and phase consistency near the flexion peaks and extension troughs. These results confirmed that the predicted

TABLE VII
JOINT-LEVEL KINEMATIC ACCURACY

Joint Angle Type	RMSE (°)↓	MAE (°)↓	R ² ↑	Corr↑	PhaseDiff (°)↓	DTW↓	SNR (dB)↓
Left knee flexion	2.61	2.01	0.97	0.99	0.72	204.20	15.83
Right knee flexion	3.69	2.82	0.96	0.98	1.12	243.90	13.53
Left ankle flexion	1.35	1.05	0.97	0.99	9.90	105.90	14.89
Right ankle flexion	1.80	1.38	0.95	0.98	9.79	121.10	13.00

TABLE VIII
QUANTITATIVE GAIT AND SYMMETRY PARAMETERS EXTRACTED

Parameter	Unit	GT	Pred	RD(%)
Stride length	mm	317.93	300.85	5.37
Cadence	Steps/min	115.71	117.39	1.45
Stance/Swing ratio	%	51.80/48.20	50.00/50.00	3.43
Symmetry Index (knee)	%	13.47	12.46	7.52
Symmetry Index (ankle)	%	118.98	110.45	7.17
PhaseDiff_mean	°	1.67	2.97	-
SNR	dB	17.95	11.42	-

key point sequence can retain the basic characteristics of motion assessment.

To quantify the accuracy of kinematic fitting, this study selected the knee and ankle joints as representative weight-bearing joints. As listed in Table VII, the RMSE of both the knee and ankle joints was less than 4°, and the correlation coefficient was close to 1, validating that the 3-D human pose estimation could support stable joint-angle calculation.

As provided in Table VIII, the key kinematic parameters were calculated from predicted 3-D key point sequences. The relative deviation (RD) of stride length was 5.37%, and the RD of cadence was 1.45%. These results demonstrated that MoViSense not only achieved stable 3-D human pose estimation but also generated interpretable kinematic parameters under sparse three-camera configurations.

Overall, the proposed framework established a pipeline from 3-D reconstruction to kinematic analysis while maintaining the stability of joint-angle curves and gait parameters, providing a practical solution for visual motion assessment.

IV. CONCLUSION

This study proposed a novel 3-D human pose estimation framework for stable pose reconstruction and interpretable kinematic analysis under sparse camera configurations. This framework integrated the MSFF module, DHF-DMA module, and BCM mechanism to improve spatial representation, enhance temporal continuity, and enforce bone-length consistency, which alleviated pose instability caused by occlusion and depth ambiguity. In addition, the MKE module generated smooth joint-angle curves and reliable kinematic parameters to support interpretable motion assessment. Comparison with state-of-the-art methods validated the superior accuracy and robustness of the proposed method under occlusions. In future

work, the study will introduce real clinical data to further improve the effectiveness and generalization of the proposed framework.

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Dezhi Zheng (Member, IEEE) received the B.S. degree in mechanical engineering and the Ph.D. degree in precision instruments and mechanical engineering from Beihang University, Beijing, China, in 2000 and 2006, respectively.

From 2006 to 2022, he was a Professor with the School of Instrumentation and Optoelectronic Engineering, Beihang University. He is currently a Professor with Beijing Institute of Technology. He is the author of more than 50 articles and more than ten inventions. His

research interests include sensor technology, machine learning, and pattern recognition.



Zhiyi Yang is currently pursuing the M.S. degree in information and communication engineering with Beijing Institute of Technology, Beijing, China.

Her research interests include signal and information processing, with a focus on the application of deep learning methods for multiview 3-D human pose estimation.



Wenfeng Liu is currently pursuing the Ph.D. degree in information and communication engineering with Beijing Institute of Technology, Beijing, China.

His research interests include machine vision-based pose measurement and dynamic sensing metrology.



Xiao Liang received the B.S. degree from Beijing Institute of Technology, Beijing, China, in 2024, where he is currently pursuing the Ph.D. degree with the Advanced Research Institute, School of Interdisciplinary Sciences, under the supervision of Prof. Dezhi Zheng.

His main research interests include gait recognition, pose estimation, and cross-view geolocalization.



Chun Hu received the Ph.D. degree in precision instruments and machinery from Beihang University, Beijing, China, in 2015.

He is currently a Professor with the School of Interdisciplinary Sciences, Beijing Institute of Technology, Beijing. His research interests include sensing mechanisms and intelligent signal processing.



Kang Ma received the B.S. degree from the School of Electrical Engineering, Yanshan University, Qinhuangdao, China, in 2016, and the M.S. degree from Beihang University, Beijing, China, in 2019.

During his Ph.D., he joined with the School of Information and Electronics, Beijing Institute of Technology, Beijing. He is currently a Post-doctoral Fellow with the School of Computer Science and Engineering, Beijing Institute of Technology. His research interests include computer vision, pattern recognition, and gait recognition.